

components and production techniques.

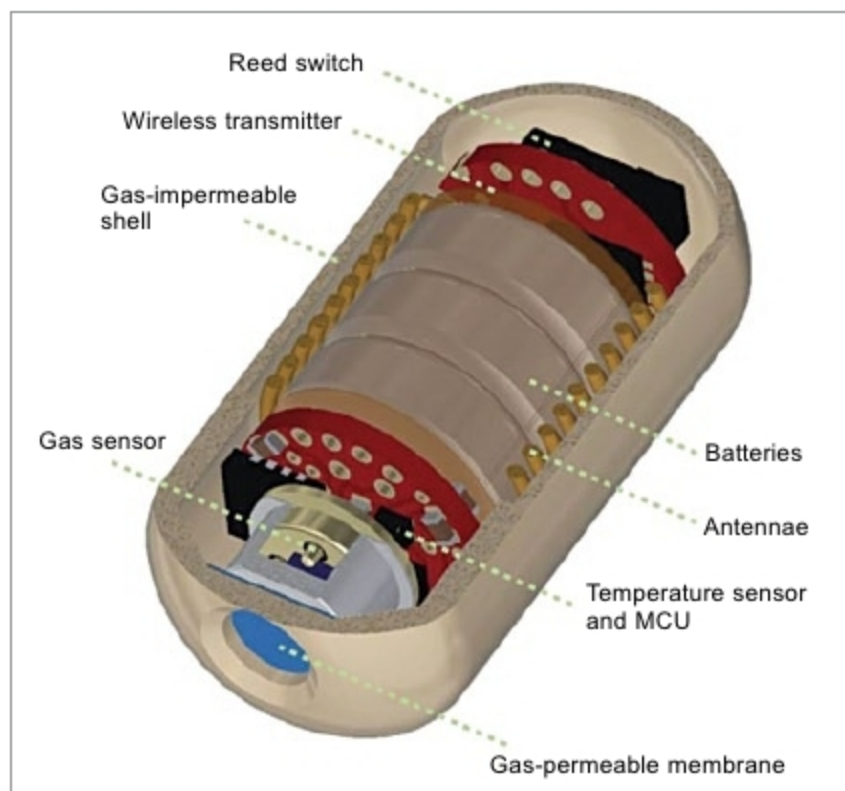
Electronic pill technology makes use of different components/parts such as drug reservoir, delivery pump, electronic microcontroller (MCU), wireless communication and sensors. These elements have to be combined in a way so as to preserve small size, reliable manufacturing and a safety profile fit for medical use. The device containing these parts is built as a small, pill-shaped capsule, which is swallowed and passed through the gastrointestinal tract.

After many years of research and development, wireless devices enclosed in capsules that can be swallowed are now hitting the market. Use of an electronic pill will free users from invasive methods such as catheters, endoscopic instruments or radioisotopes for collecting information about the digestive tract. Drug delivery using an electronic pill will also be controlled with onboard electronics, enabling precise and adaptable delivery patterns, which are not yet possible by other means.

An electronic pill has multichannel sensors that will prove to be an important tool for healthcare technology towards in-depth and detailed investigation of diseases. In addition, its uses range from drug delivery to reaching specific regions of the human body to target different types of cancer, stimulate damaged tissues, track gastric problems and measure biomarkers.

To carry out these functions, the pill is powered by an edible battery and equipped with appropriate sensors. It is important to assure that the materials used to make an edible battery are not toxic to humans, as this can cause significant complications if it gets into the digestive tract.

An electronic pill contains sensors or tiny cameras that collect information as it travels through the gastrointestinal tract before being excreted from the body a day or two later. The capsule takes measurements of the local pH and temperature inside the body.



Electronic pill structure (Credit: www.kerosi.com)

This electronic invention transmits information such as acidity, pressure and temperature levels, or images of the esophagus and intestines to the doctor's computer for analysis. Electronic pills are also being used to measure muscle contraction, ease of passage and other factors of the body to reveal information that was unavailable in the past.

In 1972, Dr John Cooper and Dr Eric Johannessen from Glasgow University (UK) developed the electronic pill. It is a medical monitoring system, measuring parameters like temperature, pH, conductivity and dissolved oxygen. It can also capture images and send it to a system.

An electronic pill has a 16mm diameter, 55mm length and 5gm weight, and can be swallowed. It is covered by a chemically-resistant polyether-terketone (PEEK) coating. As soon as the pill moves through the gastrointestinal track, it starts to detect diseases and abnormalities. The pill can easily reach areas such as small and large intestines, and deliver real-time information to an external system. Data collected is then displayed on a monitor.

A Germany-based company, Medimetrics, involved in developing oral drug delivery, has brought the idea to reality with its IntelliCap version of the electronic pill. Its electronic drug delivery pill presents an entirely new means of intelligent delivery and

monitoring. This development opens the door to a wide range of applications that are in the early stages of exploration. Taking advantage of advanced electronic monitoring and control functions, the future version of this technology is expected to be more customised and aimed at specific disease treatments.

Specifications of the electronic pill

An electronic pill consists of four microelectronic sensors. The first sensor consists of a silicon diode attached to the substrate, fabricated on two silicon chips located at the

front end of the capsule. It is used to identify body temperature. Use of a silicon-integrated circuit makes this sensor useful, and it comes at a very low cost.

The second is ion-sensitive field-effect transistor (ISFET), which is used for measuring ion concentration in solutions.

The third is direct-contact gold electrode (DCGE), which helps measure conductivity. Conductivity is measured by determining the content of water and salt absorption, and the breakdown of organic compounds into charged colloids and bile secretion.

Three-electrode electrochemical cell (TEEC) is the fourth sensor. It is used to calculate the rate of dissolved oxygen, and identify the activity of aerobic bacteria in the small and large intestines.

All these sensors are controlled by an application-specific integrated circuit (ASIC). All other components of the electronic pill, namely, 10-bit analogue-to-digital convertor (ADC), digital-to-analogue convertor (DAC), relaxation oscillator circuit (OSC) and digital signal processing circuit are connected to the ASIC consisting of analogue signal conditioning.

The circuit is powered by two SR48 silver-oxide (Ag_2O) batteries having 35 hours working capacity. Supply voltage is about 3.1V with power consumption of 15.5mW. pH and oxygen sensors are enclosed in separate 8nL